

Measuring Objectification Through the Body Inversion Paradigm: Methodological Issues

Cristina Zogmaister, Federica Durante, Silvia Mari, Franca Crippa, and Chiara Volpato

University of Milano-Bicocca

Corresponding author: Cristina Zogmaister

E-mail: cristina.zogmaister@unimib.it (CZ)

Protocol for the computation of Asymmetry Body Posture Index; ABPI

Abstract

This protocol describes the algorithm that we used to measure the asymmetry of images of body shapes. The SPSS syntax for the computation of asymmetry is provided.

The postural asymmetry of each picture is based on the following parts of the body:

1. chin
2. navel
3. top of the head
4. elbows
5. shoulders
6. hips
7. hands
8. legs

The index takes into account the Cartesian distance from the vertical line in the center of the image in the case of unique points (chin, navel, forehead). In the case of double points (elbows, shoulders, hips, hands, legs), the Cartesian distance between the right point and the mirror image of the left point is taken into consideration.

In order to compute the asymmetry index of an image, the rater has to measure the following points in pixels:

x1: top of the head;
 x2: chin;
 x3: navel;
 (x4_a, y4_a): left elbow;
 (x4_b, y4_b): right elbow;
 (x5_a, y5_a): left shoulder;
 (x5_b, y5_b): right shoulder;
 (x6_a, y6_a): left hip;
 (x6_b, y6_b): right hip;
 (x7_a, y7_a): left hand, end of middle fingertip;
 (x7_b, y7_b): right hand, end of middle fingertip;
 (x8_a, y8_a): external part of the left leg, where it touches the border of the image;
 (x8_b, y8_b): external part of the right leg, where it touches the border of the image.

Then the rater reports these positions in pixels in a spreadsheet.

The algorithm consists in the following passages:

- 1) computation of the X coordinate of the center of the body shape – which is operationalized as follows:
 - a. the center is computed for each double point, as average of the X value for the left and for the right points (e.g., the center for the elbows is computed as the average between x4_a and x4_b)
 - b. the average of the X value is computed based on the eight different parts of the body and this average is considered as the X coordinate of the center of the body shape
- 2) computation of a partial asymmetry index for each of the eight body parts:
 - a. for the three unique points, compute the values dist1, dist2, and dist3 as distances between the measured points and the center;
 - b. for the five double points, compute the X coordinate of the mirror image, mirrored on the center as computed in (a);
 - c. for the five double points, compute the values dist4, dist5, dist6, dist7, and dist8; each represents the distance between the coordinates (x_a, y_a) and (x_bspecular, y_b);
- 3) computation of the overall asymmetry index as sum of all eight distances computed in (2).

Therefore, for each image a single value (i.e., Asymmetry Body Posture Index; ABPI) is obtained that summarized the overall degree of postural asymmetry. Higher values correspond to higher levels of asymmetry.

Please find the SPSS syntax for the computation of asymmetry hereafter:

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*** SPSS syntax for the computation of the asymmetry index
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** computation of the vertical X axis - the center of the image **
COMPUTE Xcenter = mean (X1,X2,X3, mean(X4a, X4b), mean(X5a, X5b), mean(X6a, X6b), mean(X7a, X7b),
mean(X8a, X8b)).
EXECUTE.

** computation of the distance between the unique points and the center of the image **
COMPUTE dist1=abs(X1-Xcenter).
COMPUTE dist2=abs(X2-Xcenter).
COMPUTE dist3=abs(X3-Xcenter).

** computation of the specular points **
compute x4b_s = Xcenter * 2 - x4b.
compute x5b_s = Xcenter * 2 - x5b.
compute x6b_s = Xcenter * 2 - x6b.
compute x7b_s = Xcenter * 2 - x7b.
compute x8b_s = Xcenter * 2 - x8b.

** computation of cartesian distances **
compute dist4=sqrt((x4a-x4b_s)*( x4a-x4b_s)+(y4a-y4b)*(y4a-y4b)).
compute dist5=sqrt((x5a-x5b_s)*( x5a-x5b_s)+(y5a-y5b)*(y5a-y5b)).
compute dist6=sqrt((x6a-x6b_s)*( x6a-x6b_s)+(y6a-y6b)*(y6a-y6b)).
compute dist7=sqrt((x7a-x7b_s)*( x7a-x7b_s)+(y7a-y7b)*(y7a-y7b)).
compute dist8=sqrt((x8a-x8b_s)*( x8a-x8b_s)+(y8a-y8b)*(y8a-y8b)).
EXECUTE.

** computation of asymmetry as sum of the indicators
COMPUTE asymmetry=sum(dist1, dist2,dist3,dist4,dist5,dist6,dist7,dist8).
EXECUTE.
```